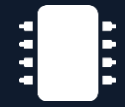


ARTIFICIAL INTELLIGENCE

MERIT BADGE









Scouting America | 4-Hour Workshop

William Mau

Today's Roadmap

0:00		Req 1 — Key Concepts & Definitions
0:30		Req 2 — AI in Everyday Life + AI or Not? Game
1:00		Req 3 — Automation Basics
1:30		Req 4 — Ethics, Games & Turing Test
2:15		**BREAK**
2:30		Req 5 — Deepfakes
2:50		Req 6 — Prompt Engineering
3:20		Req 8 — Career Exploration & Interview
3:40		Req 7b — Teachback (If Time)

REQUIREMENT 1

Key Concepts



Define 16 key AI and automation terms

AI Types — From Narrow to Super



Artificial Intelligence (AI)

Computer systems that can perform tasks that normally require human intelligence — like recognizing faces, understanding speech, or making decisions.



Narrow AI

AI designed for ONE specific task. Siri, Alexa, spam filters, and chess engines are all narrow AI. It's great at its job but can't do anything else.



General AI

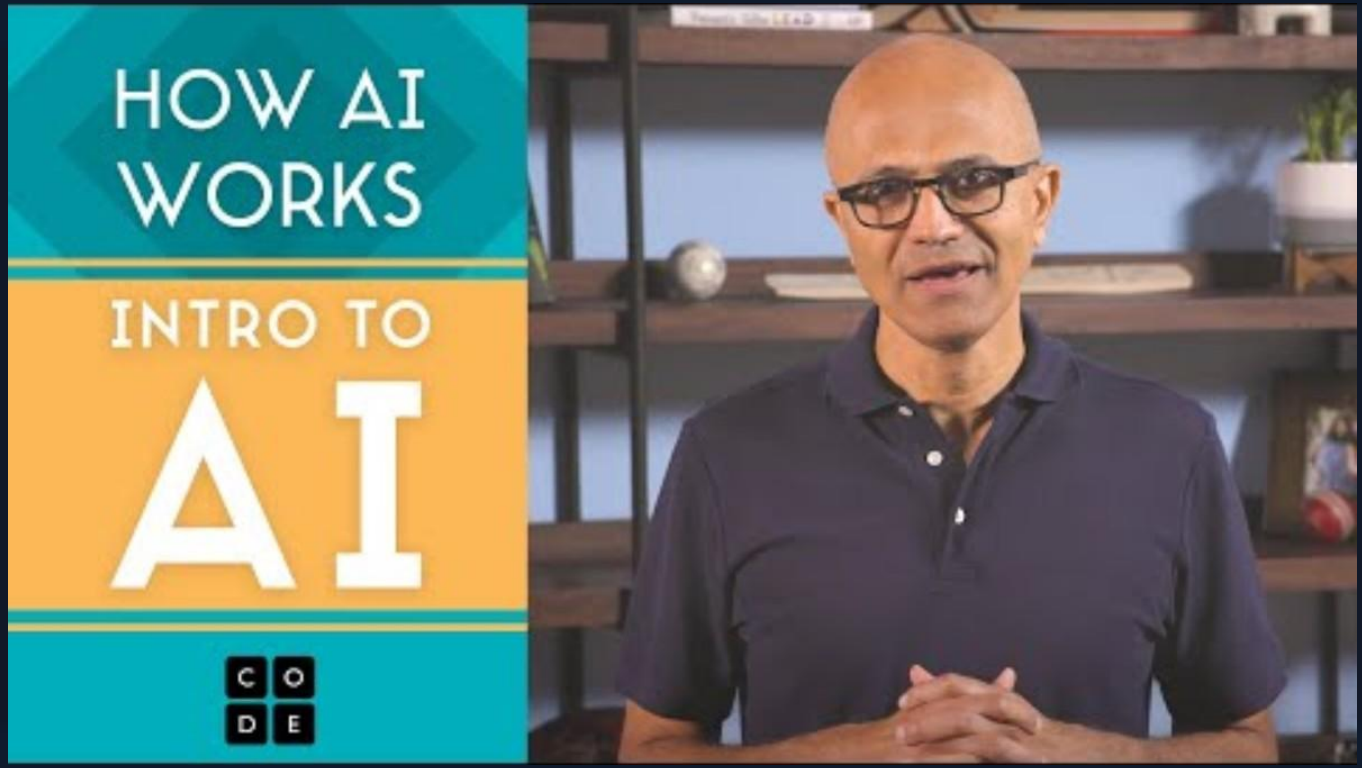
A theoretical AI that could learn and perform ANY intellectual task a human can — reasoning, creativity, emotional understanding. This doesn't exist yet. AGI



Superintelligent AI

A hypothetical AI that would surpass human intelligence in every way. This is science fiction territory — think Skynet or HAL 9000.

Video: What is AI?



Code.org — 'Introducing How AI Works' (~5 min)

Click to play video in browser

How AI Learns — Data & Machine Learning

Machine Learning (ML)

A type of AI where computers learn from examples (data) instead of being explicitly programmed. The more data, the better it gets.

Data

Information that computers collect and process — numbers, text, images, sounds. Data is the fuel that powers AI.

Databases

Organized collections of data stored electronically. Think of them like super-powered digital filing cabinets that can find anything instantly.

Variables

Named containers that store values in a program — like a label on a box. A variable called 'score' might hold the number 95.

Video: What is Machine Learning?



Code.org — 'What is Machine Learning?' (~5 min)

Click to play video in browser

AI in Action — Agents, Bots & Digital Workers

AI Agents

AI programs that can perceive their environment and take actions to achieve goals — like a self-driving car navigating traffic.

Bots

Software programs that automate tasks, often on the internet. Chatbots answer questions; web bots collect information from websites.

Digital Workers

AI-powered software that handles business tasks like processing forms, answering emails, or managing schedules — a virtual employee.

Making Things Happen — Automation & Programming

Automation

Using technology to perform tasks with minimal human involvement. Think: automatic sprinklers, dishwashers, or email auto-replies.

Tasks

Individual actions or steps in a process. Sorting emails, calculating grades, or turning on lights are all tasks.

Triggers

Events that start an automated process. Examples: a motion sensor turning on lights, or receiving an email starting a reply workflow.

Workflows

A series of tasks that happen in a specific order to complete a process. Like an assembly line for information.

Basic Programming

Writing instructions (code) that tell a computer what to do, step by step. Languages like Python and Scratch make this possible.

Key Concepts — Quick Review

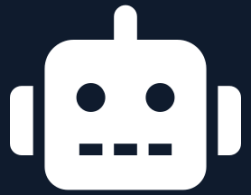
Can you explain each of these to a friend?

Artificial Intelligence (AI)	**Narrow AI**	**General AI**	**Superintelligent AI**
Machine Learning (ML)	**Data**	**Databases**	**Variables**
AI Agents	**Bots**	**Digital Workers**	**Automation**
Tasks	**Triggers**	**Workflows**	**Basic Programming**

Share one term with your neighbor — explain it in your own words!

REQUIREMENT 2

AI in Everyday Life



Identify AI all around you

ACTIVITY

10 MIN

AI Scavenger Hunt

In small groups, brainstorm examples of AI in THREE categories:

1. **10 examples in EVERYDAY LIFE**
   (apps, games, social media, smart home, etc.)

2. **5 examples in the WORKPLACE**
   (factories, offices, hospitals, stores, etc.)

3. **5 examples in SCHOOL / EDUCATION**
   (tutoring, grading, learning tools, etc.)



AI Examples — All Around Us

Everyday Life

Voice assistants (Siri, Alexa) • Netflix/Spotify recommendations • Face ID on your phone • Autocorrect & predictive text • Google Maps traffic routing • Social media feeds • Spam email filtering • Photo auto-tagging • Video game NPCs • Smart thermostats

Workplace

Medical imaging diagnosis • Fraud detection in banking • Warehouse robots • Customer service chatbots • Quality control cameras

School

AI tutoring (Khan Academy) • Plagiarism detection • Language translation • Accessibility tools • Adaptive learning apps • Duolingo



AI or Not?

10 rounds • Is it AI or just regular automation?

For each scenario, decide:
Does it use AI (learning, adapting, making decisions)?
Or is it just automation (following preset rules)?

AI OR NOT?

Round 1 of 10



Netflix recommends a movie you've never heard of — and you love it.

What do you think?

YES — That's AI!

Netflix uses machine learning to analyze your viewing history and compare it with millions of other users to predict what you'll enjoy.

AI OR NOT?



The automatic doors at the grocery store open when you walk up.

What do you think?

NOT AI — That's Automation!

Automatic doors use simple motion sensors or pressure pads — no learning, no decisions, just a trigger activating a mechanism.

AI OR NOT?

Round 3 of 10



Siri answers your question about tomorrow's weather.

What do you think?

YES — That's AI!

Voice assistants use natural language processing (AI) to understand your speech, interpret meaning, and generate a response.

AI OR NOT?



Your home thermostat turns on the heat every day at 7 AM.

What do you think?

NOT AI — That's Automation!

A basic timer-based thermostat follows a preset schedule — it doesn't learn or adapt. (A Nest thermostat WOULD be AI!)

AI OR NOT?

Round 5 of 10



Gmail moves a suspicious email to your spam folder.

What do you think?

YES — That's AI!

Spam filters use machine learning to analyze patterns in millions of emails, learning to identify new types of spam over time.

AI OR NOT?



A traffic light changes from red to green every 60 seconds.

What do you think?

NOT AI — That's Automation!

A fixed-timer traffic light follows a simple program — no analysis or learning involved. (Smart traffic lights with cameras ARE AI!)

AI OR NOT?



A Tesla navigates itself through city traffic.

What do you think?

YES — That's AI!

Self-driving cars use computer vision, deep learning, and sensor fusion to understand their environment and make real-time driving decisions.

AI OR NOT?



A calculator solves 847 × 293 instantly.

What do you think?

NOT AI — That's Automation!

Calculators follow fixed mathematical rules — they don't learn, adapt, or make decisions. They're powerful tools, but not AI.

AI OR NOT?



Your phone unlocks when it recognizes your face.

What do you think?

YES — That's AI!

Face ID uses neural networks trained on thousands of facial features. It can recognize you in different lighting, with glasses, or with a new haircut.

AI OR NOT?

Round 10 of 10



A vending machine gives you change after you insert $5.

What do you think?

NOT AI — That's Automation!

The vending machine follows simple math rules — subtract item price from amount inserted, dispense the difference. No learning needed.

ACTIVITY

5 MIN

AI Timeline Challenge

Create a timeline with 5 key milestones in AI history.

Some ideas to research:

- 1950 — Alan Turing proposes the Turing Test
- 1956 — 'Artificial Intelligence' coined at Dartmouth
- 1997 — IBM Deep Blue beats world chess champion
- 2011 — IBM Watson wins Jeopardy!
- 2016 — Google AlphaGo beats world Go champion
- 2022 — ChatGPT launches, changes everything



REQUIREMENT 3

Automation Basics



Understand how automation works around us

ACTIVITY

8 MIN

Automation Around Us

Same groups — now brainstorm AUTOMATION examples:

1. **10 examples in EVERYDAY LIFE**
2. **5 examples in the WORKPLACE**
3. **5 examples in SCHOOL / EDUCATION**

Remember: Automation = performing tasks WITHOUT human involvement, but NOT necessarily learning or adapting.



Automation Examples — All Around Us

Everyday Life

Dishwashers • Automatic sprinklers • Alarm clocks • Auto-pay bills • Cruise control • Automatic car wash • Robotic vacuum (basic) • Coffee maker timer • Garage door opener • Washing machine cycles

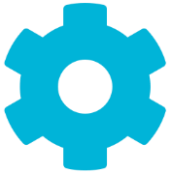
Workplace

Assembly line robots • Barcode scanners • Auto email responses • Inventory reorder systems • Payroll direct deposit

School

Auto-grading (multiple choice) • Bell schedules • Automatic backups • Timed test submissions • Attendance scanners

How Automation Works



Handles Repetitive Tasks

Automation performs the same action over and over without getting tired or bored. It frees humans to focus on creative, complex work.



Reduces Errors

Machines follow precise instructions every time. No typos, no miscalculations, no 'oops I forgot.' Consistency is automation's superpower.



Optimizes Resources

Automation uses time, energy, and materials more efficiently. Smart irrigation waters plants only when needed — saving water and money.

ACTIVITY

5 MIN

Automation Timeline Challenge

Create a timeline with 5 milestones in automation history.

Some ideas:

- 1784 — First automatic flour mill (Oliver Evans)
- 1913 — Ford's moving assembly line
- 1947 — First automatic transmission in cars
- 1961 — First industrial robot (Unimate)
- 1975 — First programmable home appliances
- 2002 — Roomba robotic vacuum launched



REQUIREMENT 4

Ethics in AI



Explore bias, privacy, and responsible AI use

PREREQ SHARE

What Did You Discover?

You researched ethical concerns in AI — now let's share!

Bias

What examples of AI bias did you find? How can biased data lead to unfair outcomes?

Privacy

How does AI collect and use your personal data? What concerned you most?

AI Decision-Making

Should AI make decisions about people's lives? Where do you draw the line?

Each Scout: share one thing that surprised you from your research

Video: Training Data & Bias



Code.org — 'AI: Training Data & Bias' (~5 min)

Click to play video in browser



What Would You Do?

5 ethical scenarios • No right or wrong answers

Think about what's right, what's fair, and what a Scout would do. Be ready to explain your reasoning!

WHAT WOULD YOU DO?



Your friend uses ChatGPT to write their entire history essay and submits it as their own work.

A. It's fine — it's just a tool, like using a calculator

B. It's cheating — they didn't learn anything

C. It depends — using AI to help outline or research is OK, but submitting AI writing as your own isn't

Discuss with your group — there's no single right answer!

WHAT WOULD YOU DO?



A company's AI hiring tool automatically rejects applicants from certain zip codes. The zip codes happen to be low-income neighborhoods.

A. The AI is just using data — it's not intentionally biased

B. This is unfair discrimination, even if unintentional

C. The company should test for bias and fix the algorithm

Discuss with your group — there's no single right answer!

WHAT WOULD YOU DO?



Your school wants to use AI cameras to monitor hallways and flag 'suspicious behavior' to prevent bullying.

A. Great idea — if it stops bullying, it's worth it

B. Too invasive — students deserve privacy at school

C. Only OK with strict rules about what gets flagged and who sees it

Discuss with your group — there's no single right answer!

WHAT WOULD YOU DO?



An AI-generated artwork wins first place in an art competition, beating human artists who spent months on their pieces.

A. Fair game — the person who created the prompt showed creativity

B. Unfair — AI art should be in a separate category

C. AI is a tool like Photoshop — it depends on how much human input was involved

Discuss with your group — there's no single right answer!

WHAT WOULD YOU DO?



A social media app uses AI to show you content that makes you angry because angry users spend more time on the app.

A. That's their business — I can just close the app

B. This should be illegal — they're manipulating emotions for profit

C. Users should be able to see and control how the algorithm works

Discuss with your group — there's no single right answer!

ACTIVITY | 10 MIN

Your AI Ethics Code

Write your own personal ethical guidelines for using AI.

Think about:

- When is it OK to use AI for schoolwork? When isn't it?
- What personal information should AI never have?
- How should you treat AI-generated content?
- What responsibility do AI creators have?

Write at least 5 rules you'll follow. Be specific!



The Turing Test — Can Machines Think?

What is the Turing Test?

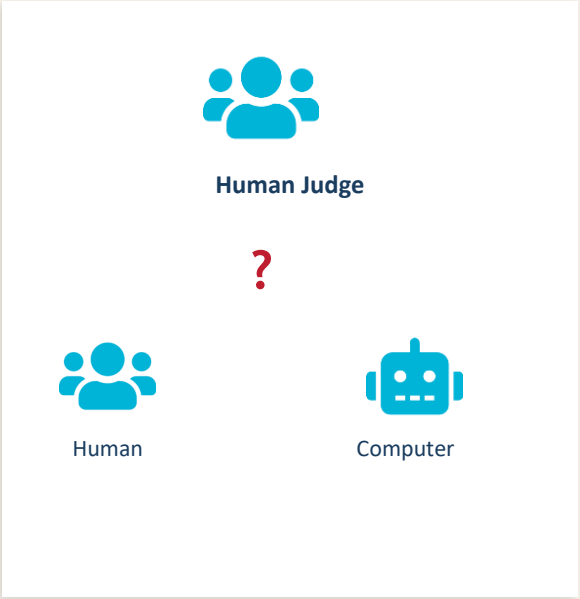
Proposed by Alan Turing in 1950, the Turing Test asks: **Can a machine fool a human into thinking it's also human?**

How it works:

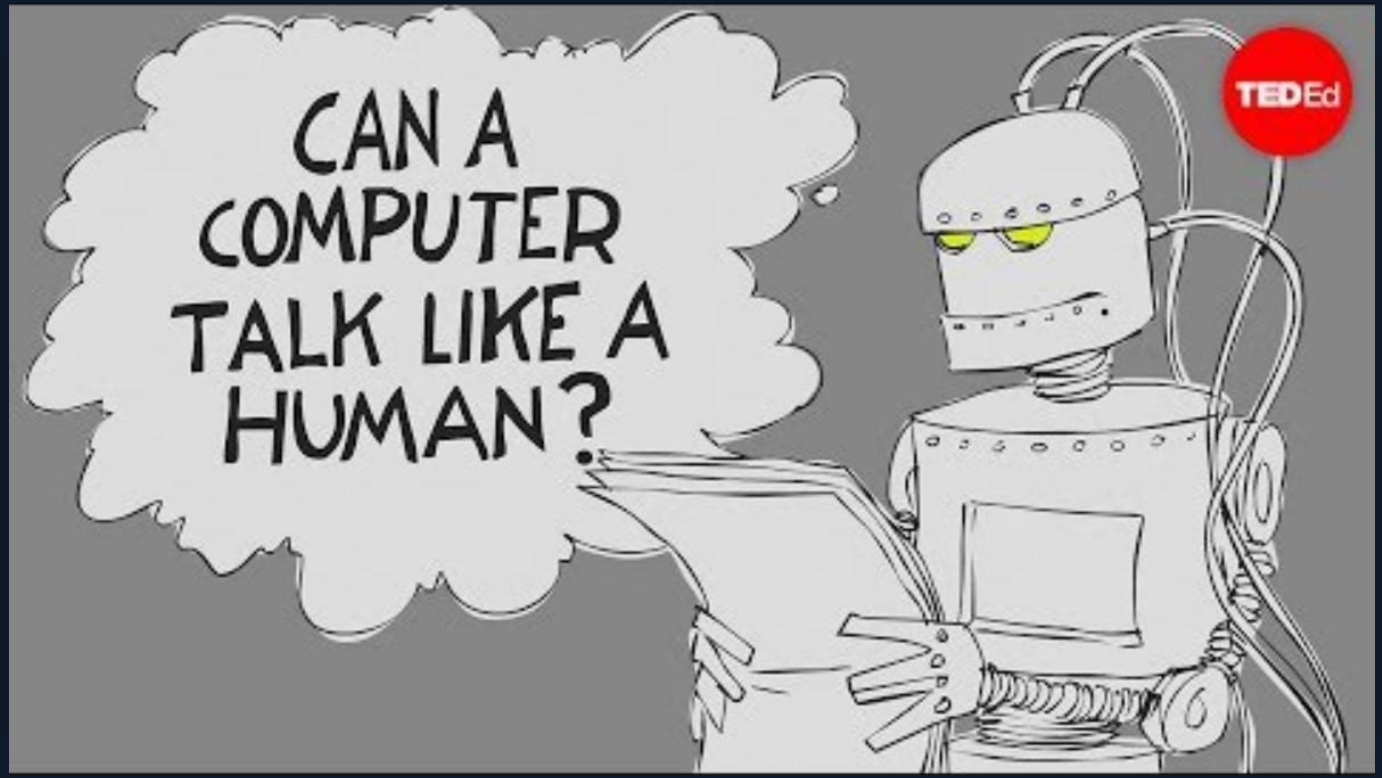
A human judge has text conversations with both a human and a computer. If the judge can't reliably tell which is which, the computer passes the test.

Why it matters:

It was one of the first serious attempts to define machine intelligence. Today, chatbots like ChatGPT are making us rethink what 'passing' really means.



Video: The Turing Test



TED-Ed — 'The Turing Test: Can a Computer Pass for a Human?' (~4 min)

Click to play video in browser



BREAK TIME

15 minutes — Be back at the posted time!

Stretch, grab a snack, and recharge!

REQUIREMENT 5

Deepfakes



Understand the threat and how to respond

What is a Deepfake?

Deepfake = AI-generated fake media that makes someone appear to say or do something they never did. The name combines **'deep learning'** (a type of AI) with **'fake.'**

Face Swapping

Putting one person's face onto another's body in video

Voice Cloning

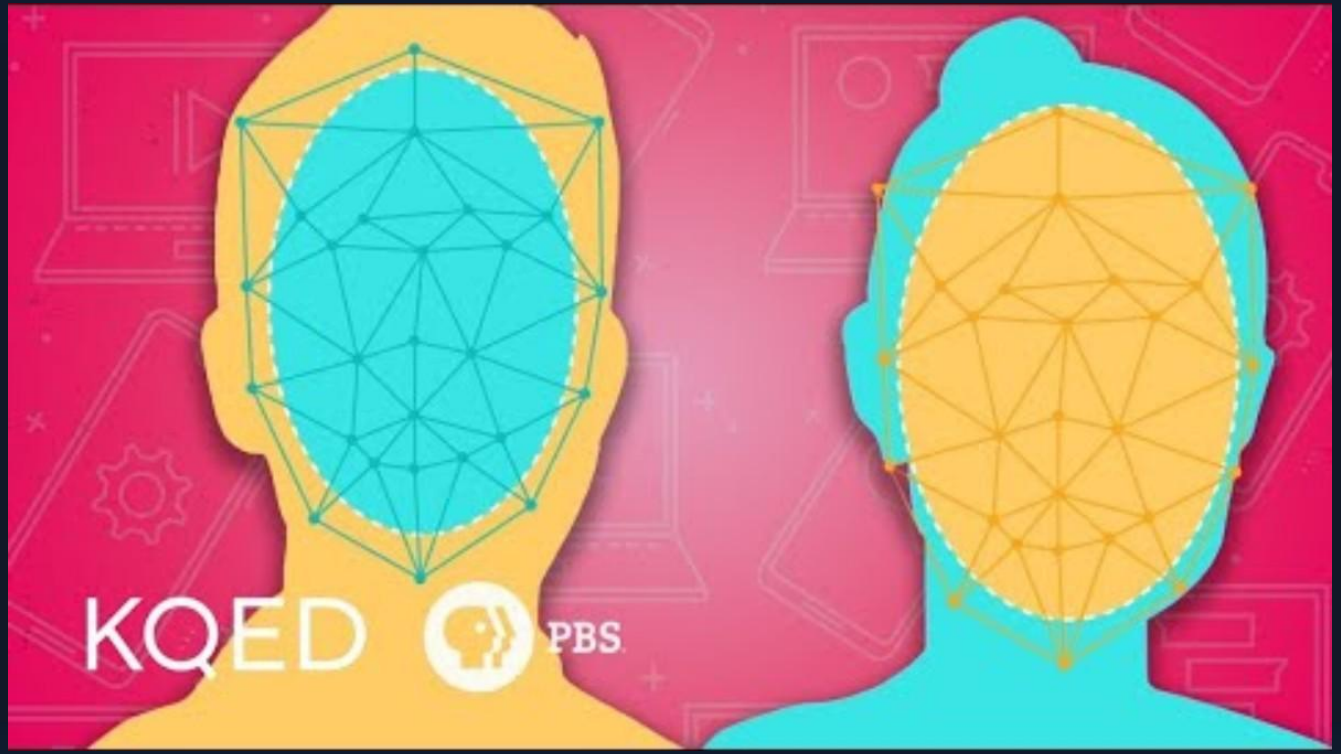
Creating a synthetic copy of someone's voice from audio samples

Text Generation

AI writing text that mimics a specific person's style

By 2025, an estimated 8 million deepfakes were shared on social media — a 900% annual growth rate.

Video: Can You Spot a Deepfake?



KQED Above the Noise — 'Deepfakes: Can You Spot a Phony Video?' (~7 min)

Click to play video in browser

How Deepfakes Impact People



Reputation Damage

A fake video can destroy someone's reputation overnight — even after it's proven fake, many people will have already seen and believed it.



Emotional Harm

Victims of deepfakes experience anxiety, depression, and a feeling of losing control over their own identity and image.



Misinformation

Deepfakes can spread false information about events, elections, or public figures — undermining trust in real media.



Financial Fraud

Voice-cloned deepfakes have been used to impersonate executives and authorize fraudulent bank transfers worth millions.

What To Do If You Encounter a Deepfake

1

Don't share it

Stop the spread. Even sharing to debunk it gives it more views.

2

Verify the source

Check if reputable news outlets have covered it. Reverse image search.

3

Look for red flags

Unnatural blinking, weird lighting, audio sync issues, warped edges.

4

Report it

Use the platform's reporting tools. Flag as manipulated media.

5

Support the victim

If someone you know is targeted, be supportive. Help them report it.

6

Tell a trusted adult

If a deepfake targets you or a friend, tell a parent, teacher, or counselor immediately.

REQUIREMENT 6

Developing AI Skills



Learn to communicate effectively with AI

How AI Learns — And Its Limits

How AI Learns

1. Collects massive amounts of data
2. Finds patterns in that data
3. Builds a model based on patterns
4. Tests predictions against real results
5. Adjusts and improves over time

AI Limitations

- No true understanding — it mimics patterns
- Can 'hallucinate' — make up false info confidently
- Only as good as its training data
- Struggles with common sense reasoning
- Can't feel emotions or have experiences
- Doesn't know what it doesn't know

Video: How Chatbots & LLMs Work (Optional)



Code.org — 'How Chatbots and Large Language Models Work' (~5 min)

Click to play video in browser

5 Methods for Communicating with AI

1

Be Specific

Instead of 'tell me about dogs,' try 'explain 3 differences between golden retrievers and labrador retrievers for a school report.'

2

Give Context

Tell the AI who you are and what you need. 'I'm a 7th grader writing a science report' helps AI tailor its response.

3

Set Constraints

Define format, length, and style. 'Write a 3-paragraph summary using simple language' gives clear boundaries.

4

Iterate & Refine

Treat it as a conversation. If the first response isn't right, follow up: 'Make it shorter' or 'Add more examples.'

5

Verify Outputs

Always check AI responses against reliable sources. AI can sound confident while being completely wrong.

What is Prompt Engineering?

Prompt Engineering is the skill of crafting clear, effective instructions for AI to get the best possible output. Think of it as learning to speak AI's language — the better your prompt, the better the result.

Bad Prompt

"Tell me about space"

Result: A vague, unfocused response that could be about anything from astronomy to rocket design to sci-fi movies.

Great Prompt

"Explain 3 reasons why Mars is hard to colonize, for an 8th grade science presentation. Use simple language and include one surprising fact."

Result: Focused, grade-appropriate, specific, and includes exactly what you asked for.

Live Demo: 3 School-Related AI Tasks

LIVE DEMO

1 Research Helper

"I'm writing a 5-paragraph essay on the water cycle for 7th grade science. Create an outline with a thesis statement, 3 body paragraph topics, and a conclusion summary."

Why it works:
Specific grade, format, and task

2 Study Guide Creator

"Create 10 multiple-choice quiz questions about the American Revolution for an 8th grade history test. Include answer keys and brief explanations."

Why it works:
Clear format, quantity, subject, and level

3 Writing Feedback

"Review this paragraph and suggest improvements for clarity, grammar, and stronger word choices. Explain each suggestion so I can learn from it."

Why it works:
Asks for teaching, not just fixing

Why Prompt Clarity Matters

Clear communication with AI is a skill that transfers everywhere — school assignments, job applications, giving instructions to teammates, and even talking to adults about what you need.



School

Write better essays, ask teachers clearer questions, give more effective presentations



Work

Give clearer instructions to coworkers, write professional emails, delegate tasks effectively



Everyday Life

Explain problems to tech support, give directions, communicate needs more precisely

REQUIREMENT 8

Career Exploration



Discover AI careers and interview a professional

AI Careers — Where Could You Work?

Machine Learning Engineer

Tech / Software

Builds and trains AI models. Median salary: ~$130K+

Data Scientist

Analytics

Analyzes data to find insights and patterns. Works in every industry.

AI Ethics Researcher

Policy / Research

Ensures AI is fair, transparent, and doesn't harm people.

Robotics Engineer

Engineering

Designs and programs robots for manufacturing, healthcare, and exploration.

NLP Specialist

Linguistics / CS

Teaches computers to understand human language (like Siri or ChatGPT).

AI Product Manager

Business / Tech

Guides AI product development. Bridges tech teams and business goals.

Video: AI, Privacy & the Future of Work (Optional)



Code.org — 'Ethics & AI: Privacy & the Future of Work' (~5 min)

Click to play video in browser

ACTIVITY | 15 MIN

Interview a Professional

Your counselor works in the tech industry. This is your chance to ask real questions!

Sample Interview Questions:

1. What does a typical day look like in your job?
2. How do you use AI or automation in your work?
3. What challenges do you face working with AI technology?
4. What education or training did you need to get your job?
5. What do you think AI will look like in 10 years?
6. What advice would you give a Scout interested in tech careers?

Come up with at least 2 of your own questions too!

IF TIME ALLOWS



Teachback Challenge

REQUIREMENT 7b

Pick ONE topic you learned today and teach it to the group in 5 minutes.

Include: a clear explanation, a real-world example, and why it matters.

If we don't have time, you can complete Req 7a or 7b on your own for full credit.

Teachback Topic Ideas

- What is Machine Learning?
- How Face Recognition Works
- AI Bias — Why It Matters
- Deepfakes & How to Spot Them
- Narrow AI vs. General AI
- How Self-Driving Cars Work
- Prompt Engineering Tips
- The Turing Test
- AI in Healthcare
- Automation in Everyday Life
- AI Ethics — My Guidelines
- The History of AI

Pick a topic, prep for 3 minutes, then teach for 5 minutes!

Badge Completion Checklist

Req 1: Key Concepts — Defined 16 terms	COMPLETE
Req 2: AI Basics — Identified examples + AI or Not? game + timeline	COMPLETE
Req 3: Automation Basics — Identified examples + timeline	COMPLETE
Req 4: Ethics — Shared research, What Would You Do?, ethics code, Turing Test	COMPLETE
Req 5: Deepfakes — Explained what they are and how to respond	COMPLETE
Req 6: Developing AI Skills — Communication methods + prompt engineering	COMPLETE
Req 7: Practical Application — Teachback (complete if time allowed)	IF TIME
Req 8: Career Exploration — Interviewed a professional	COMPLETE

If you didn't finish Req 7 today, you can complete 7a or 7b on your own to earn the badge.





Congratulations!

You've completed the AI Merit Badge workshop!

Remember: AI is a powerful tool — use it wisely, ethically, and creatively.

The future of AI will be shaped by people like YOU.